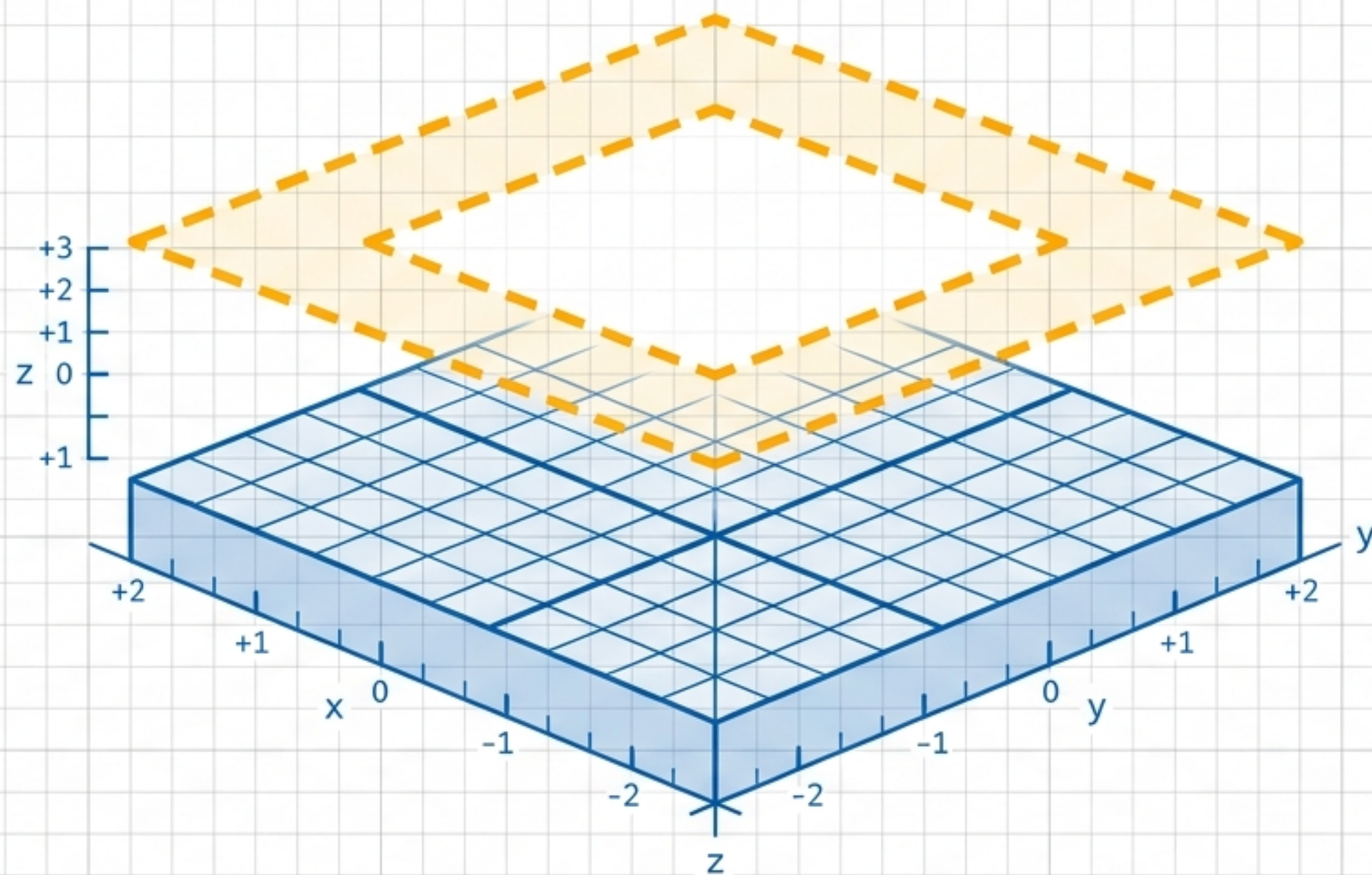


[afflom/F1 :: v0.13.0]



The $\mathbb{F}_1$ Square: A Formal Research Base for the Riemann Hypothesis
Formalizing the missing surface $\text{Spec } \mathbb{Z} \times_{\mathbb{F}_1} \text{Spec } \mathbb{Z}$ in Lean 4.

The Riemann Hypothesis is OPEN. This is its Research Base.

WHAT IT IS

- A precise formulation of the $\mathbb{F}_1$ construction problem.
- A choice-free Lean 4 formalization of the exact mathematics surrounding the Riemann Hypothesis (RH).
- A system that mechanically verifies the honest epistemic status of every piece.

WHAT IT IS NOT

- A claim to have solved the Riemann Hypothesis.
- An assertion that the central crux is constructed.
- A repository of unverified, assumed, or hidden proofs.

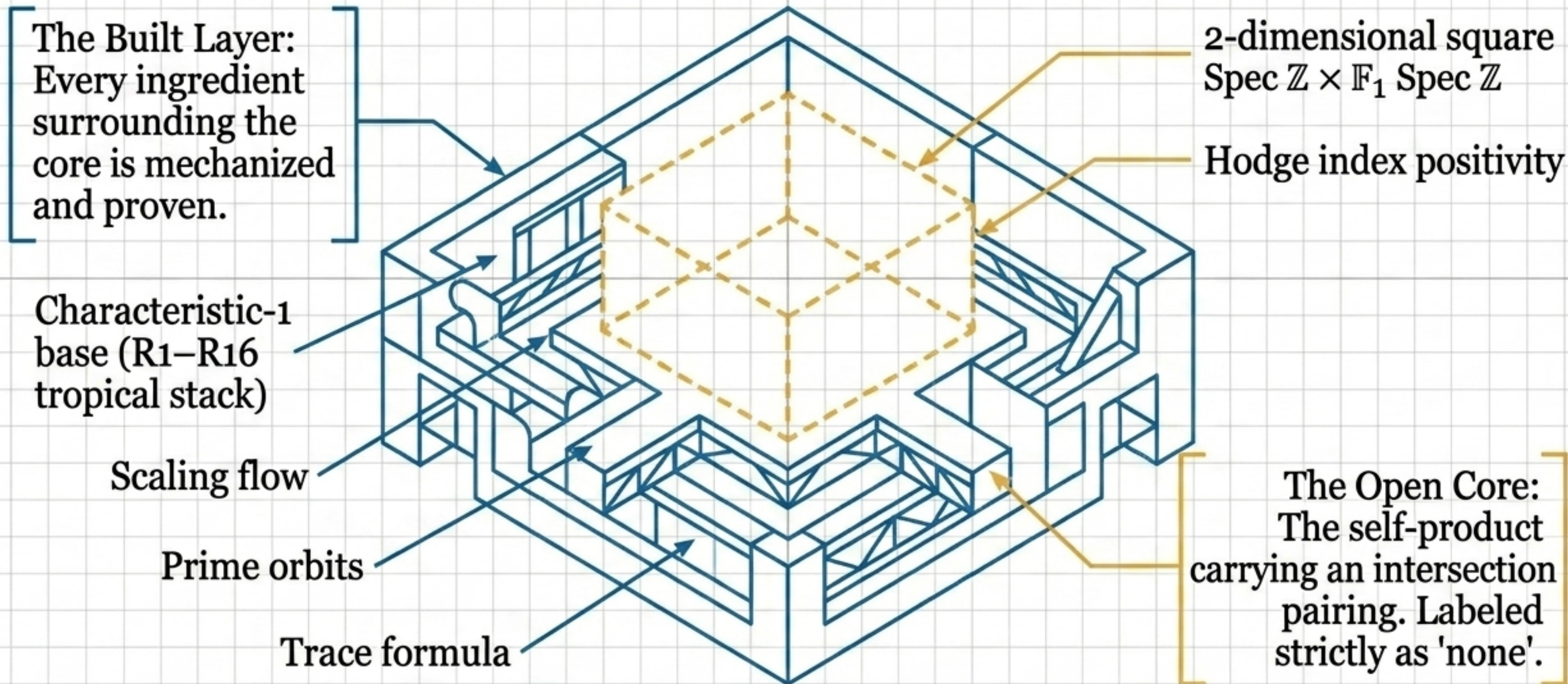
[The crux is never asserted as proven anywhere in this repository.]

The Missing Analog: Function Fields vs. $\mathbb{Q}$

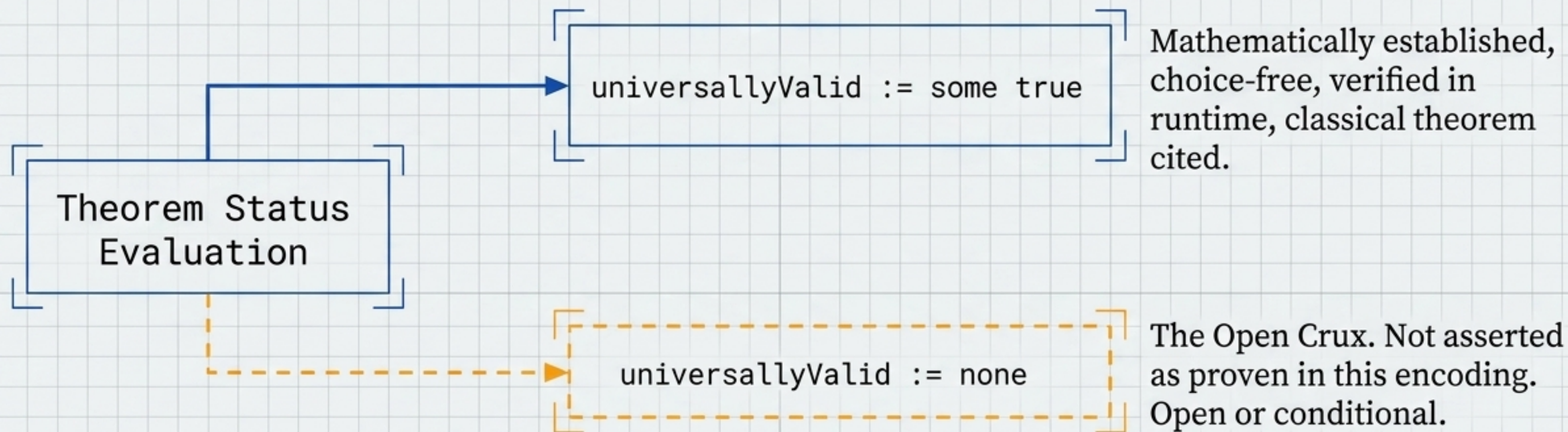
Function Field over $\mathbb{F}_q$ (Solved)		Number Field over $\mathbb{Q}$ (Open)
Setting	Curve over $\mathbb{F}_q$	$\text{Spec } \mathbb{Z}$ over $\mathbb{F}_1$
Mechanism	Frobenius scaling flow	Built Scaling Flow ✓
Cohomology	Étale cohomology of $\mathbb{C} \times \mathbb{C}$	Unconstructed cohomology
The crux	Hodge Index Theorem (Proven positivity confining eigenvalues)	Intersection-positivity (MISSING)

This negative-definiteness is the Riemann Hypothesis.

Framing the Unbuilt Crux



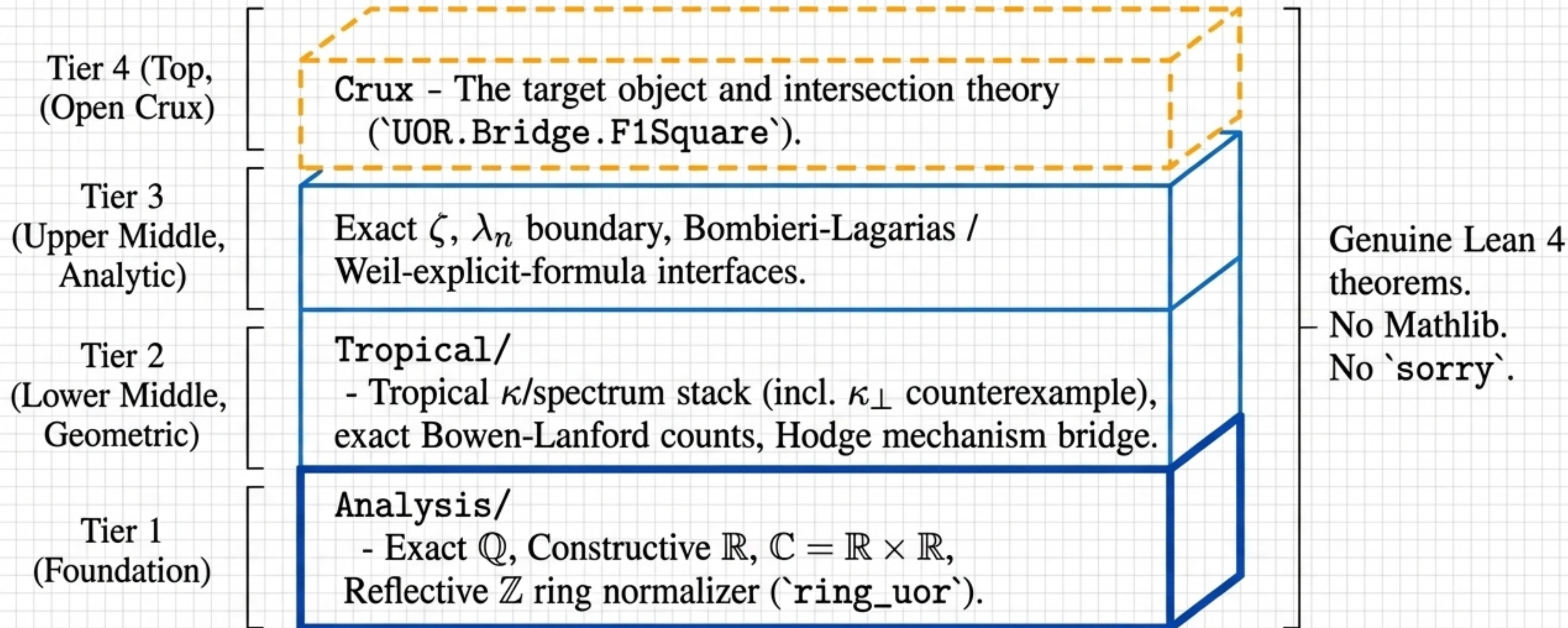
Absolute Honesty as Code



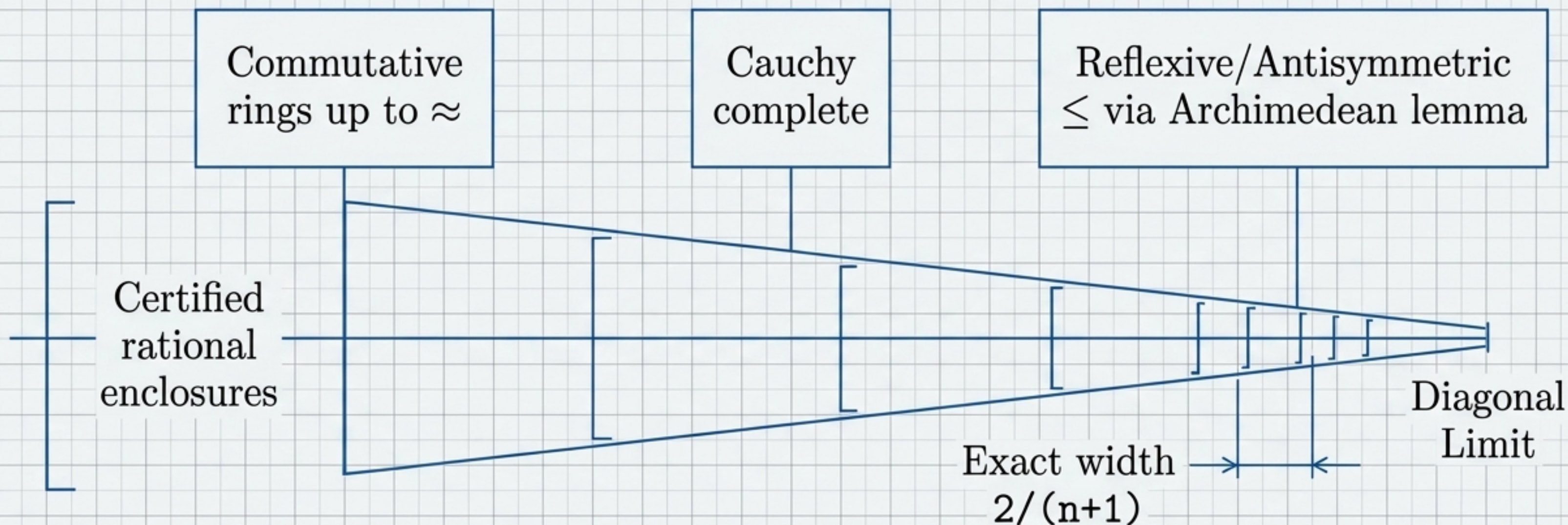
“No field asserts an unproven claim as true. If the crux is ever genuinely proved, its status becomes `some true`. Until then, the system enforces humility as a verifier, not a prohibition.”

The Lean 4 Proof Layer Architecture

F1Square/



Building Constructive Reals Without Choice



Real powers, division R_{div} , and reciprocals R_{inv} (using positivity-as-data) are built choice-free, culminating in the diagonal limit.

The Transcendentals Arc (v0.13.0)

$$\text{Exp/Euler } (e) = \sum \frac{1}{i!}$$

Everywhere-defined: Diagonal of rational partial sums $S_{Rj}(x_{Rj})$, rigorously bounded by geometric truncation, uniform Lipschitz, and factorial-growth estimates.

$$\frac{\text{Cos}}{\text{Sin}} = \frac{\sum (-x^2)^n}{(2n + \text{off})!}$$

The alternating diagonal dominated by $\exp(M^2)$.

$$\text{Log}(x) = 2 \cdot \text{artanh} \left(\frac{x-1}{x+1} \right)$$

Built via the Möbius t-map. Uses genuine positivity-as-data: a witness derives the rational modulus $1/M \leq x \leq M$.

Pinning the Analytic Boundary: Li's Criterion

1 Encoded open (`none`) as `'LiPositive'`.

$$\text{RH} \Leftrightarrow \lambda_n > 0 \quad \forall n \geq 1$$

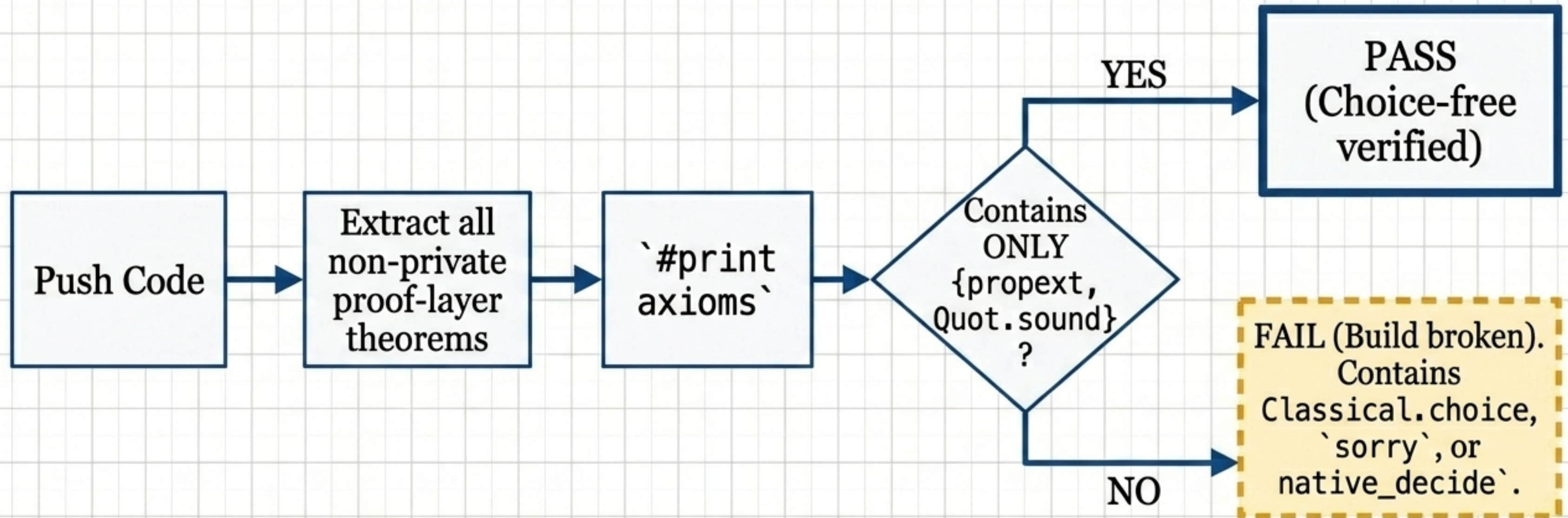
$$\left[\zeta(s) = \sum \frac{1}{i^s} \text{ (for } s \geq 2) \right]$$

3 Shipped as exact-bounded objects
(certified rational enclosures).

2 Features a finite-check guard
(`LiPositive` = $\bigwedge$ all finite truncations).
No `'decide'` constitutes a proof.

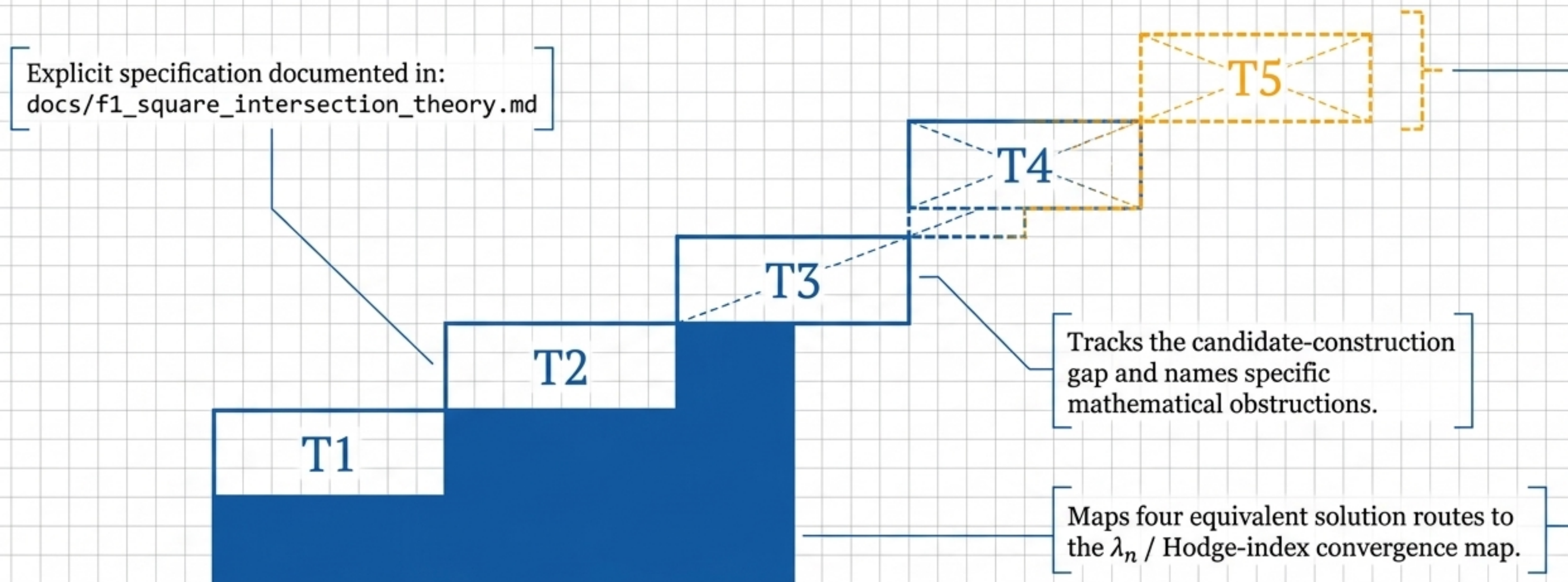
The continuation into the critical strip is deliberately deferred, not fabricated. We state the genuine ζ -derived Li sequence faithfully.

Self-Enforcing Rigor via Continuous Integration



The two surviving axioms are forced by core internals and are constructively uncontroversial. No `sorry`. No `native\_decide`. No stray axioms.

The T1–T5 Verification Ladder



The architecture is not just a statement of ignorance; it is a meticulously documented coordinate system for a future proof.

Preserved and Pinned for the Future

GitHub Release Sync

`v0.13.0`

`lake-manifest.json`

Strict dependency pinning to
UOR-Framework `v0.5.2` for
absolute Lean 4 reproducibility.

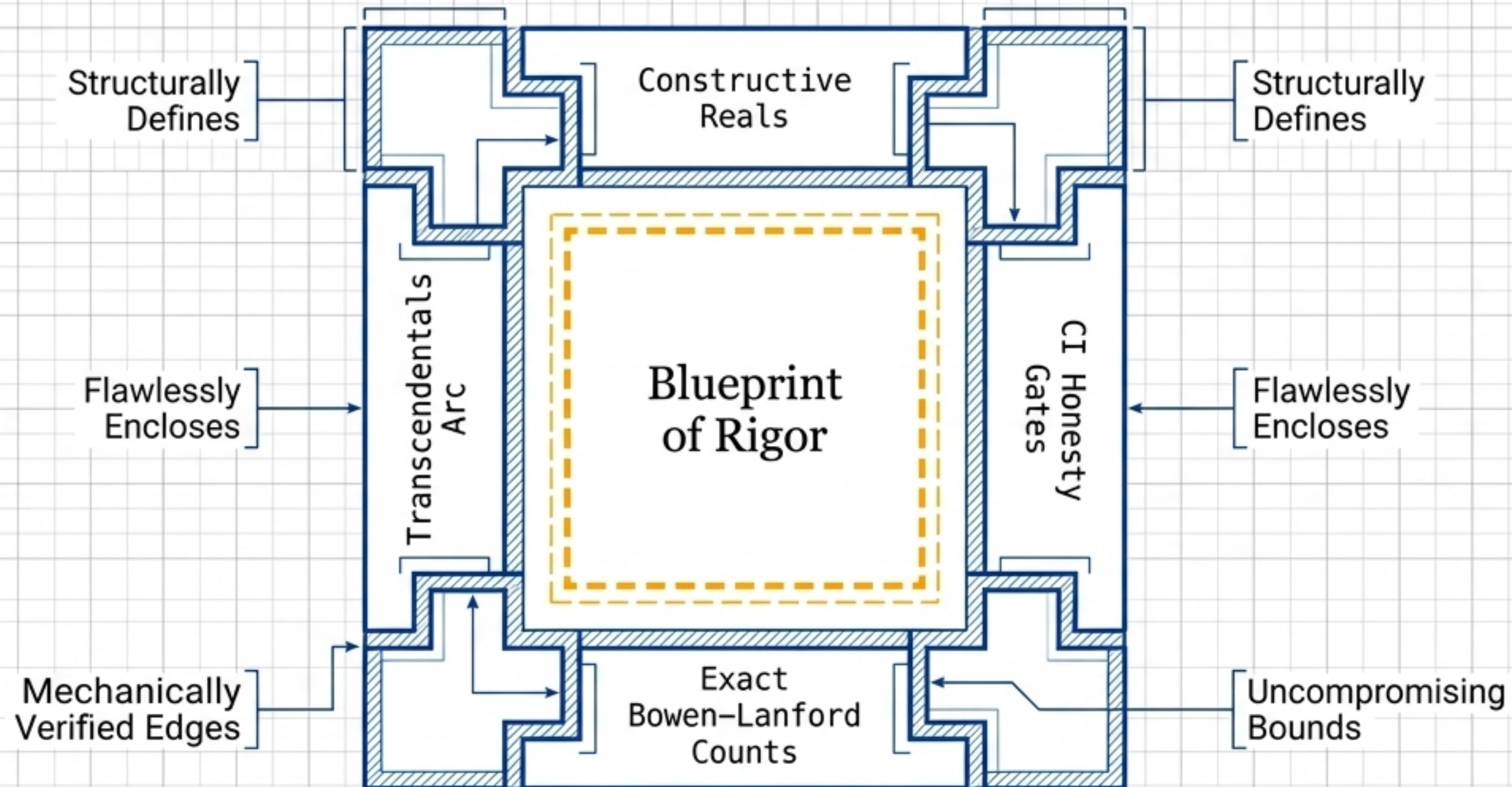
Zenodo DOI Minting

CITATION.cff

GitHub
Release
Trigger

Permanent
DOI Minted

The Perfect Crucible for the Hardest Problem



The `afflom/F1` repository does not solve the Riemann Hypothesis. Instead, it engineers an uncompromising, choice-free, mechanically verified crucible. The structure is flawless. The space is mathematically honest. Only the crux remains.